

AI SKIN DISEASE DETECTION SYSTEM USING MULTI-MODAL AI

Complete Project Design, Implementation, and Evaluation Report

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1. Project Abstract & Objectives

1.1 Abstract

DermaScan AI is a deployable, clinical-grade multi-modal skin disease detection system designed to classify 18 distinct skin conditions. The core methodology fuses visual features extracted from skin lesion photographs using a deep Convolutional Neural Network (EfficientNet-B0) with linguistic features extracted from patient symptom descriptions (supporting English, Hindi, and Kannada) via a lightweight offline Count Vectorizer projection encoder. To allow zero-downtime serverless hosting, the PyTorch model checkpoints are converted to lightweight Open Neural Network Exchange (ONNX) runtimes. In addition, an Out-of-Distribution (OOD) pipeline rejects non-skin inputs, a dynamic ReportLab PDF compiler issues downloadable reports, and geolocations fetch nearest dermatologists via OpenStreetMap. The system achieves 63.95% accuracy on an isolated, 100% real clinical test set of 18 classes under 5 epochs.

2. System Performance

2.1 Evaluation Summary

Metric	Validation Set	Isolated Test Set
Accuracy	55.81%	63.95%
Precision	64.07%	59.62%
Recall	55.81%	63.95%
F1-Score	52.98%	58.76%